Technology Stack

Date	1 July 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Jinja2 templates rendered by Flask)	A simple, lightweight form-based UI (soil nutrients + weather inputs) was enough for the use case; Jinja2 templates let the backend render index.html and result.html directly without needing a separate JS framework.
2	Backend / Server-Side	Python, Flask	Flask is lightweight and well-suited for serving a scikit-learn model behind a simple REST-style /predict endpoint; Python keeps the ML model and the web server in the same language, simplifying development.
3	Database / Data Storage	None (trained model stored as serialized .pkl files)	The application doesn't need to persist transactional data; the trained model (crop_model.pkl) and label encoder (label_encoder.pkl) are loaded directly via pickle at startup, keeping the app stateless and fast.

4	Cloud / Hosting / Deployment	Vercel (serverless, @vercel/python builder), Gunicorn for WSGI	Vercel provides free, easy serverless hosting with automatic scaling for a low-traffic academic project; vercel.json routes all requests to api/index.py, avoiding the need to manage dedicated server infrastructure.
5	Version Control & CI/CD	GitHub	GitHub is the standard pairing with Vercel deployments, enabling automatic redeployment on every push and easy collaboration among team members.
6	Third-Party APIs / Other Tools	scikit-learn, NumPy, Pandas (ML/data libraries)	scikit-learn trains and serves the crop-recommendation model; NumPy structures the incoming form data (N, P, K, temperature, humidity, pH, rainfall) into the array format the model expects; Pandas was used during data preprocessing/training.